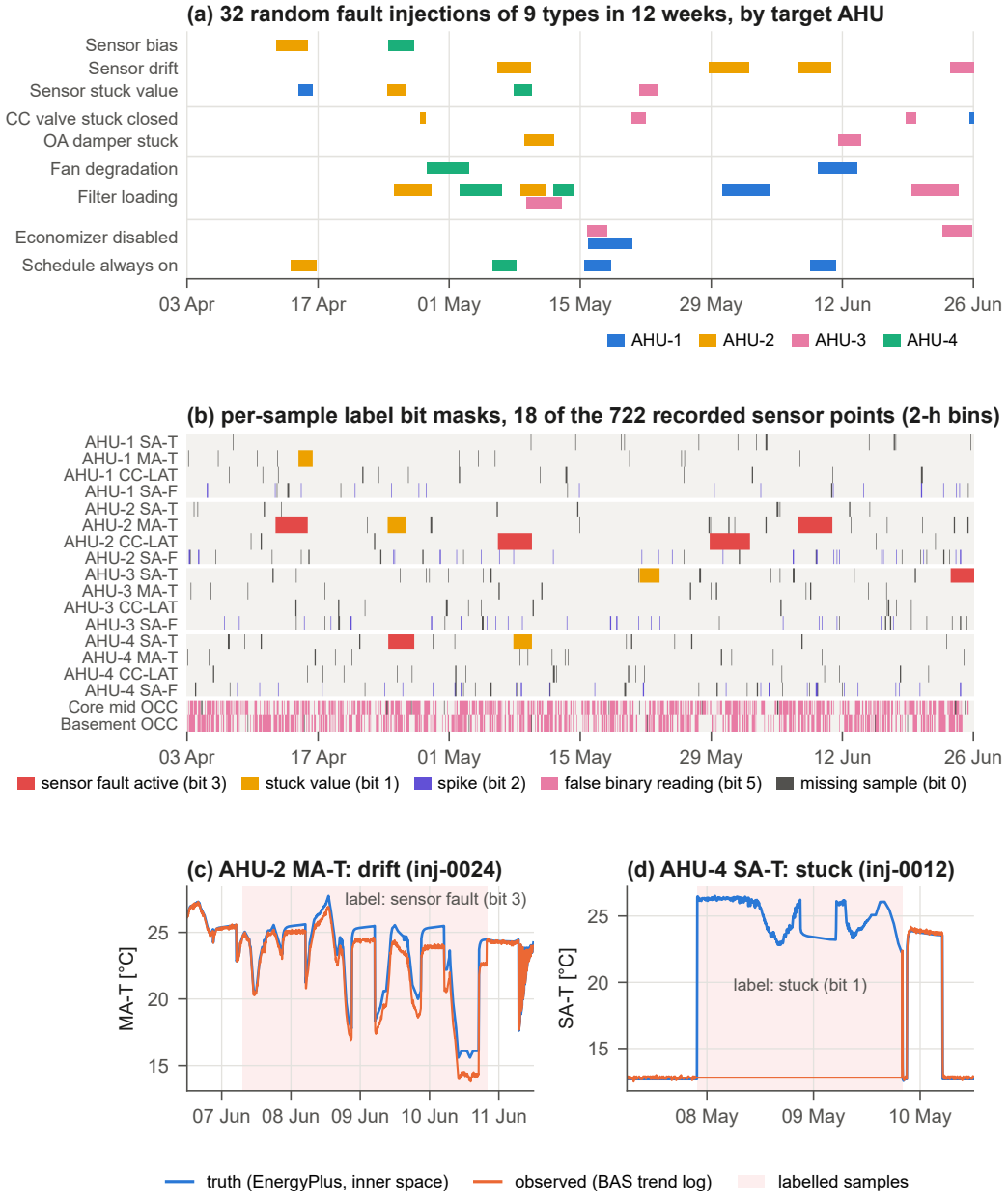

(a) 32 random fault injections of 9 types in 12 weeks, by target AHU
Sensor bias
Sensor drift
Sensor stuck value
CC valve stuck closed
OA damper stuck
Fan degradation
Filter loading
Economizer disabled
Schedule always on
03 Apr
17 Apr
01 May
15 May
29 May
12 Jun
26 Jun
AHU-1
AHU-2
AHU-3
AHU-4
(b) per-sample label bit masks, 18 of the 722 recorded sensor points (2-h bins)
AHU-1 SA-T
AHU-1 MA-T
AHU-1 CC-LAT
AHU-1 SA-F
AHU-2 SA-T
AHU-2 MA-T
AHU-2 CC-LAT
AHU-2 SA-F
AHU-3 SA-T
AHU-3 MA-T
AHU-3 CC-LAT
AHU-3 SA-F
AHU-4 SA-T
AHU-4 MA-T
AHU-4 CC-LAT
AHU-4 SA-F
Core mid OCC
Basement OCC
03 Apr
17 Apr
01 May
15 May
29 May
12 Jun
26 Jun
sensor fault active (bit 3)
stuck value (bit 1)
spike (bit 2)
false binary reading (bit 5)
missing sample (bit 0)
(c) AHU-2 MA-T: drift (inj-0024)
label: sensor fault (bit 3)
MA-T [°C]
25
20
15
07 Jun
08 Jun
09 Jun
10 Jun
11 Jun
(d) AHU-4 SA-T: stuck (inj-0012)
label: stuck (bit 1)
SA-T [°C]
25
20
15
08 May
09 May
10 May
truth (EnergyPlus, inner space)
observed (BAS trend log)
labelled samples